

2.1. Using hockey stick decomposition or otherwise find the sums

$$\sum_{k=1}^n 6k^2 + 2k.$$

2.2 Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{4/3}{4}, \quad \binom{-1/2}{4}.$$

3.1. Using Gauss summation trick or otherwise find the sum

$$-3\binom{n}{0} - 1\binom{n}{1} + 1\binom{n}{2} + 3\binom{n}{3} + \cdots + (2n-3)\binom{n}{n}.$$

3.2. Consider the equation $a + b + c = 11$ with integers $a \geq 3, b \geq 2, c \geq 2$.

(a) How many solutions are there?

(b) Which solution would you encode as $\star\star \mid \star\star\star$?

4.1. Solve the recurrence equation:

$$b_n = \frac{n+3}{n+4}b_{n-1}, \quad b_1 = 10.$$

4.2. Here L is the lag operator and $x_n = 6^n$. Find the values

$$(1-6L)x_n, \quad (1-2L)x_n, \quad (1-6L)(1-3L)x_n.$$

5.1. Consider the recurrence equation $y_n - 7y_{n-1} + 12y_{n-2} = 0$.

Find all solutions of the form $y_n = \lambda^n$.

5.2. You know that $r_n = c \cdot 2^n + d \cdot n \cdot 3^n + n$ is the solution of some recurrence equation with initial conditions $r_0 = 3$ and $r_1 = 12$.

Find c and d .

6.1. Calculate the following complex values:

$$(6-3i) \cdot (2+4i), \quad \frac{1+8i}{3+4i}, \quad (1-5i)^3.$$

6.2. Consider the recurrence equation

$$a_n - 4a_{n-1} + 5a_{n-2} = 4n - 14.$$

Find the solution of the form $a_n = \alpha n + \beta$.

7.1. Extract the sequence (a_n) from the generating functions:

$$g(t) = \frac{1}{1-5t}, \quad g(t) = \frac{1}{1+2t}, \quad g(t) = \frac{t}{1-3t}.$$

7.2. Decompose the fraction into a sum of more simple fractions

$$g(t) = \frac{5-5t}{(1+2t)(1-3t)}.$$

- 10.1 I throw a fair coin until the sequence HHT appears. Draw the directed graph for this game.
- 10.2 Alice and Bob throw a fair coin until the sequence $A = HHH$ or the sequence $B = THT$ appears. Alice wins if the sequence A appears first, Bob wins if the sequence B appears first. Draw the directed graph for this game.
- 11.1 Write the cyclic monomial that corresponds to the transposition $(123)(4)(5)(6789)$.
- 11.2 You know that the cycle index of the group G that acts on the colourings X is given by

$$CI(z_1, z_2, z_3, z_4) = \frac{1}{24}(z_1^6 + 6z_2z_4 + 8z_3^2 + 9z_1^2z_2^2)$$

- a) How many different colourings in 1, 2 or 3 colours are possible?
- b) How many different colourings in exactly 3 colours are possible?

You don't need to finalize all arithmetic calculations in the quiz, just write the answer as an expression with numbers.

- 12 Let's consider a 4×4 mosaic of gems where each gem can be one of k types. Consider a 180° clockwise rotation of the mosaic.
- 12.1 Represent this rotation as a sequence of cycles and write the corresponding cycle monomial.
- 12.2 How many 4×4 mosaics are fixed by this rotation?
- 13.1 Let e be some edge of the complete graph K_4 .
Draw the graphs K_4 , $K_4 - e$ and K_4/e .
- 13.2 Compute the chromatic polynomial $\text{chrom}(k, C_4)$ where C_4 is the cycle with 4 vertices.
- 14.1 Let e be some edge of the complete multipartite graph $K_{1,3}$.
Draw the graphs $K_{1,3}$, $K_{1,3} - e$ and $K_{1,3}/e$.
- 14.2 Compute the chromatic polynomial of the following graph:

